

METASPLOITABLE 2 -BASED PROJECT REPORT

Project: Metasploitable 2 & Mutillidae II Setup and Configuration

1. Overview

This report documents the complete step-by-step setup of Metasploitable 2 and OWASP Mutillidae II vulnerable

web application on VMware Workstation in Windows. The lab is intended strictly for ethical hacking and security

practice.

2. Lab Environment- Host OS: Windows- Virtualization: VMware Workstation- Target VM: Metasploitable 2 (Ubuntu 8.04)- Web App: OWASP Mutillidae II- Network Mode: Host-only / NAT (isolated lab)

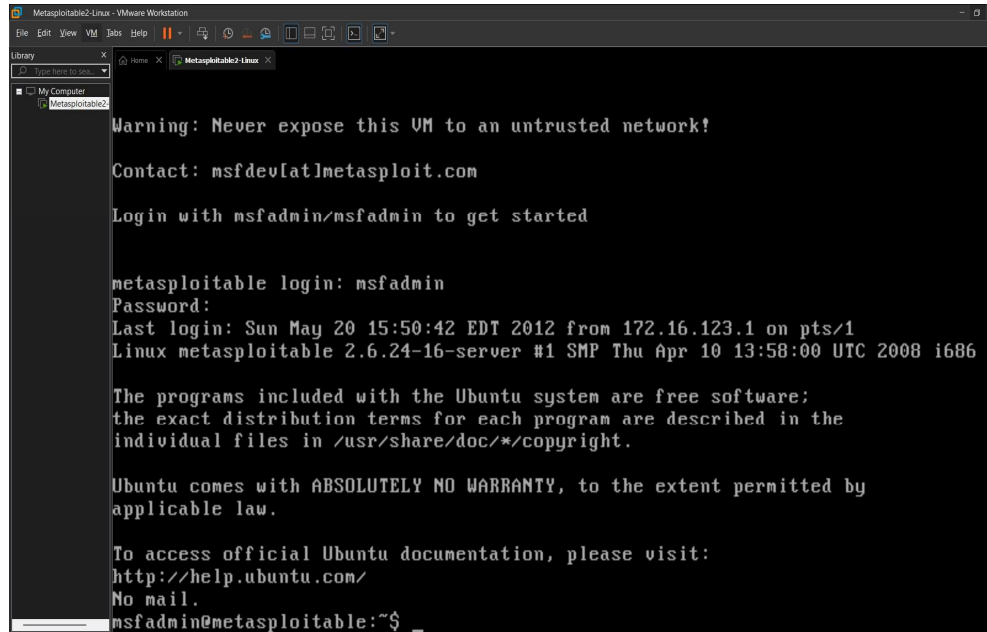
3. Step-by-Step Setup Process- Downloaded and imported Metasploitable 2 VM into VMware Workstation.- Booted the VM and logged in successfully.- Created and logged in with a custom user account (harshit_13).- Started Apache and MySQL services using init scripts.- Navigated to /var/www/mutillidae directory.- Executed Mutillidae database setup script (set-up-database.php).- Verified database creation without errors.- Accessed Mutillidae II from browser using VM IP address.

4. Commands Used

```
sudo su
/etc/init.d/apache2 start
/etc/init.d/mysql start
cd /var/www/mutillidae
php set-up-database.php
chmod -R 777 /var/www/mutillidae
ifconfig
```

Step 1: Login to Metasploitable 2

The Metasploitable 2 virtual machine was booted in VMware Workstation. The default credentials were used to log in successfully, confirming the VM was running properly.



```
Metasploitable2-Linux - VMware Workstation
File Edit View VM Jats Help
Library
Type here to search
My Computer
Metasploitable2

Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

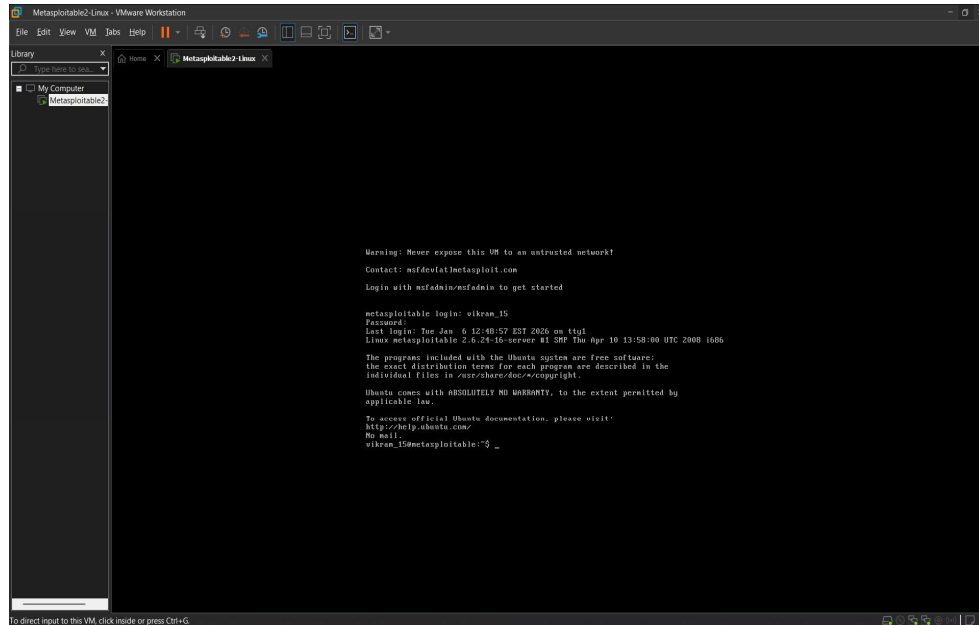
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ _
```

Step 2: Switching to Root User

To start system services like Apache and MySQL, root privileges are required. The user switched to the root account using the su command.



The screenshot shows a VMware Workstation window titled "Metasploitable2 Linux - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tools, Help), a toolbar, and a left sidebar with a "Library" pane showing "My Computer" and "Metasploitable2". The main terminal window displays the following text:

```
Warning: Never expose this VM to an untrusted network!
Contact: msfdev@metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: vikram_15
Password:
Last login: Tue Jan 6 12:48:57 EST 2026 on ttty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

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individual files in /usr/share/doc/*/*-copyright.

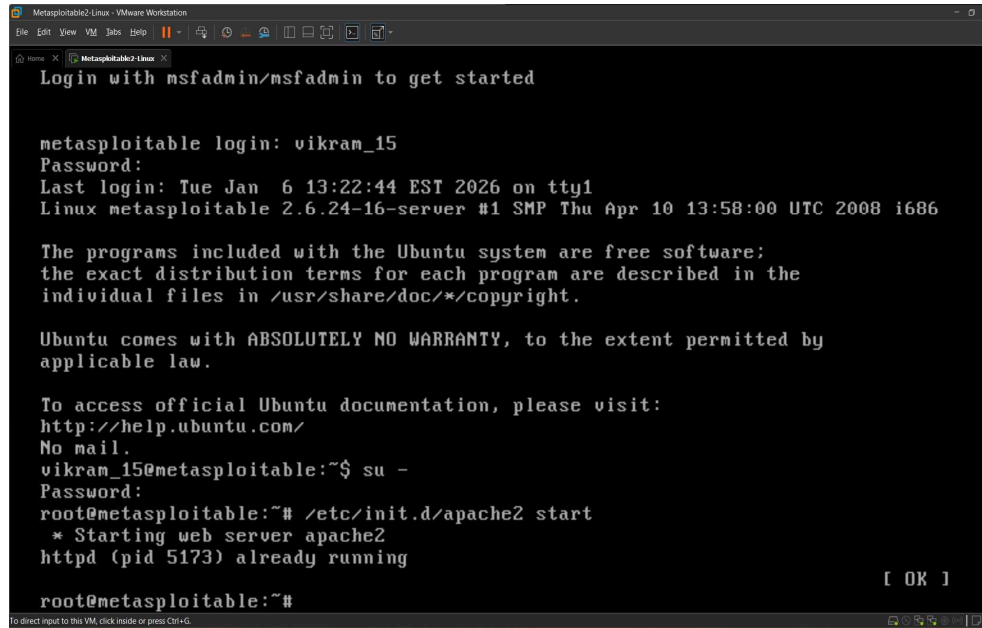
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
vikram_15@metasploitable:~$ _
```

At the bottom of the window, a status bar reads: "To direct input to this VM, click inside or press Ctrl+G."

Step 3: Starting Apache Web Server

Apache web server was started using legacy init scripts. The output confirms that Apache is running and bound to port 80.



```
Metasploitable2 Linux - VMware Workstation
File Edit View VM Jobs Help
Metasploitable2 Linux
Login with msfadmin/msfadmin to get started

metasploitable login: vikram_15
Password:
Last login: Tue Jan 6 13:22:44 EST 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

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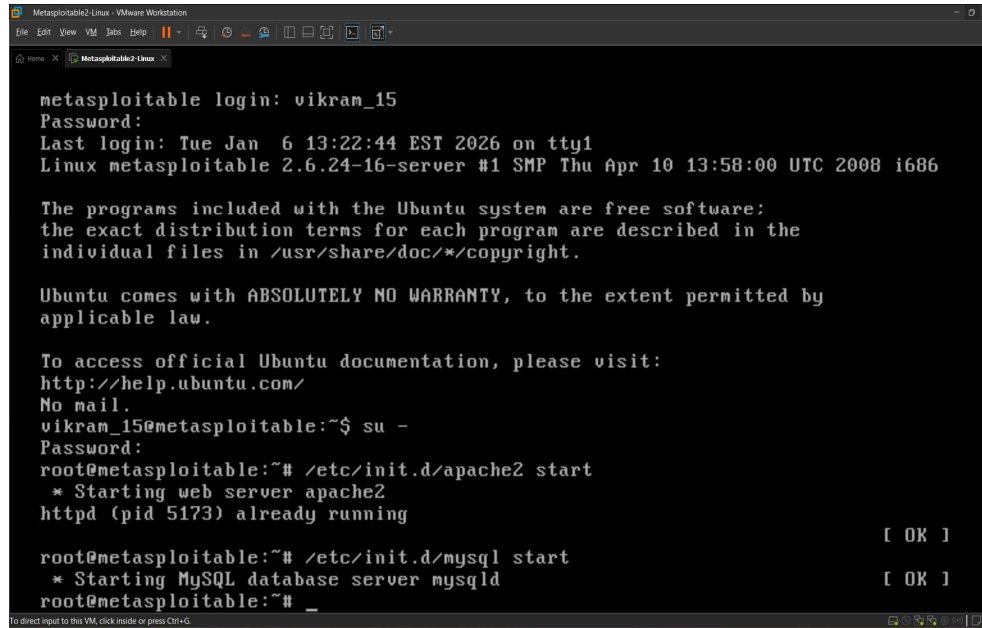
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
vikram_15@metasploitable:~$ su -
Password:
root@metasploitable:~# /etc/init.d/apache2 start
* Starting web server apache2
httpd (pid 5173) already running

[ OK ]

root@metasploitable:~#
```

Step 4: Starting MySQL Database Server

The MySQL database server was started successfully. This is required for Mutillidae II to store and retrieve data.



```
Metasploitable2 Linux - VMware Workstation
File Edit View VM Jobs Help
Metasploitable2 Linux x

metasploitable login: vikram_15
Password:
Last login: Tue Jan  6 13:22:44 EST 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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applicable law.

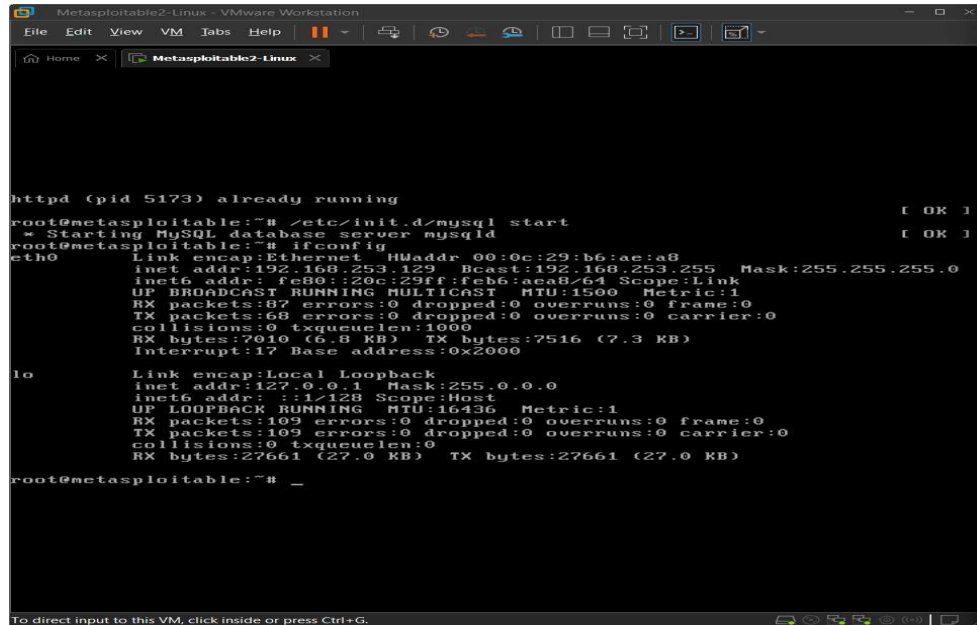
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
vikram_15@metasploitable:~$ su -
Password:
root@metasploitable:~# /etc/init.d/apache2 start
* Starting web server apache2
httpd (pid 5173) already running                                     [ OK ]

root@metasploitable:~# /etc/init.d/mysql start
* Starting MySQL database server mysqld                             [ OK ]
root@metasploitable:~# _
```

To direct input to this VM, click inside or press Ctrl+G.

Step 5: Identifying IP Address

The ifconfig command was used to identify the IP address assigned to the Metasploitable VM, which is required to access the application from a browser.



```
Metasploitable2 - Linux - VMware Workstation
File Edit View VM Tabs Help
Metasploitable2 - Linux

httpd (pid 5173) already running
root@metasploitable:~# /etc/init.d/mysql start
* Starting MySQL database server mysqld
root@metasploitable:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:b6:ae:a8
          inet addr:192.168.253.129  Bcast:192.168.253.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:feb6:aea8/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:87 errors:0 dropped:0 overruns:0 frame:0
          TX packets:68 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:7910 (6.6 KB)  TX bytes:7516 (7.3 KB)
          Interrupt:17 Base address:0x2000

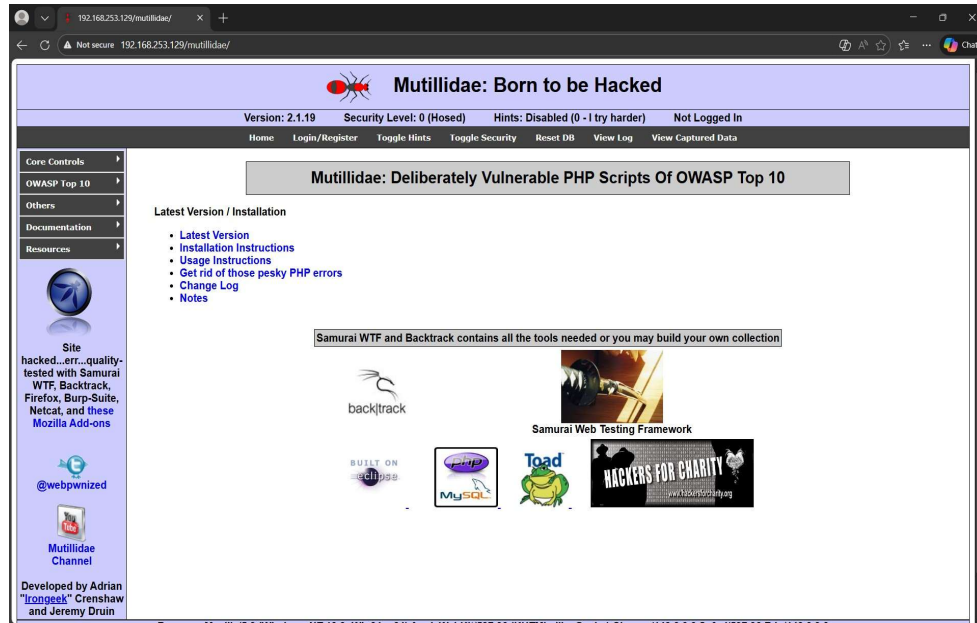
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:109 errors:0 dropped:0 overruns:0 frame:0
          TX packets:109 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27661 (27.0 KB)  TX bytes:27661 (27.0 KB)

root@metasploitable:~# _
```

To direct input to this VM, click inside or press Ctrl+G.

Step 6: Accessing Mutillidae II in Browser

Using the identified IP address, Mutillidae II was accessed from the host browser. The successful page load confirms proper server and database configuration.



Conclusion

The screenshot-based steps confirm the successful installation and configuration of Metasploitable 2 and Mutillidae II. This lab environment can now be used for practicing OWASP Top 10 web application vulnerabilities.